

Babies' First Bandits

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Outline

- The basic multi-armed bandits (MAB) model
- Algorithms and proof ideas
- Extensions

Example - Slot Machines



Example - Slot Machines

- There are 20 slot machines.
 - Each requires \$1 to play.
 - Has an unknown payoff.
- You have \$1000.
- How should you choose a sequence of machines to maximize total reward?



Example - Slot Machines

- Random.
- Play each machine 50 times.
- Play each machine 25 times, choose the best for the remaining 500 rounds.
- Play each machine 10 times, eliminate the worst few, repeat.



Examples - Ad Placements

- In each round a customer arrives.
- Arms: ads.
- Reward: 1 if customer clicks on ad, 0 otherwise.

Examples - Routing

- In each round a packet needs to be sent from source to destination.
- Arms: paths.
- Reward: quality of path, e.g., negative latency.

Examples - Recommender Systems

- In each round a customer arrives.
- Arms: content (e.g., movies, posts, etc.).
- Reward: interaction signals (e.g., click, rating, etc.).

Examples - Medical Trials

- In each round a patient arrives.
- Arms: medical treatments.
- Reward: treatment effectiveness.

Stochastic Bandits - Model

- Formally, there is a time horizon T and a fixed set of K arms $[K]$.
- In each round $t \in [T]$:
 - Learner chooses an arm $a_t \in [K]$.
 - Learner earns reward $r_t \sim D_{a_t}$.



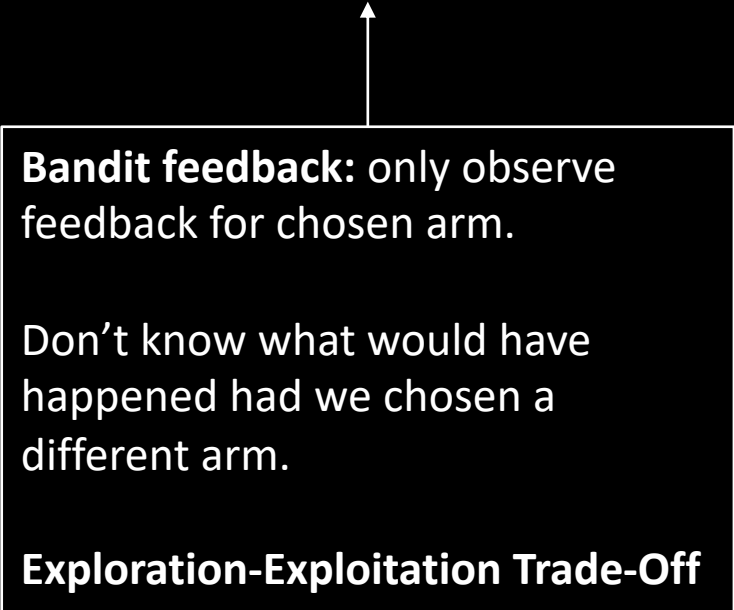
Each arm a has a reward distribution with mean $\mu(a)$.

The reward in round t is sampled from the reward distribution of arm a_t .

Distributions are unknown to the learner!!!

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Bandit feedback: only observe feedback for chosen arm.

Don't know what would have happened had we chosen a different arm.

Exploration-Exploitation Trade-Off

Stochastic Bandits - Goal

- Goal: maximize total expected reward

$$E \left[\sum_{t=1}^T \mu(a_t) \right].$$

Expectation with respect
to randomness in the
choice of a_t .

Stochastic Bandits - Regret

- We measure the performance of an algorithm A by regret.

$$R_T(A) = OPT - E \left[\sum_{t=1}^T \mu(a_t) \right].$$

Total expected reward of
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Total expected reward of
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- **Goal: minimize regret.**
- We want regret as a sublinear function of T .

- This implies $\frac{R_T(A)}{T} \rightarrow 0$ as $T \rightarrow \infty$.

On average, algorithm's
performance $\rightarrow$ OPT.

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 - Just pick the arm with the highest mean reward.
- But we don't know the true means.
 - Use empirical means as an estimate for the true means.
 - With high probability, these will be close to each other.
 - Hoeffding: after N samples, with prob. $\geq 1 - 2T^{-4}$.

$$|\mu(a) - \bar{\mu}(a)| \leq \sqrt{\frac{2 \log T}{N}}$$

Algorithm - Failed Attempt

- In each round, choose arm with highest empirical mean.
- Regret: $O(T)$.
 - One arm has reward 0.4 deterministically.
 - The other is Bernoulli with parameter 0.9.
 - With probability 0.1, we always choose the suboptimal arm.

Algorithm - Explore-Then-Commit

- Play each arm N times.
- Let $\hat{a}$ be an arm with the highest empirical mean.
- In all future rounds, play $\hat{a}$.



Algorithm - Explore-Then-Commit

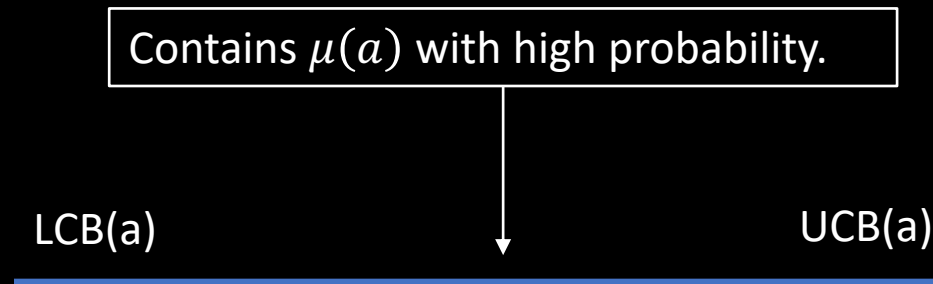


- How should we choose N ?
 - If N is large, empirical means are good estimates.
 - But we also incur regret till we identify the best arm.
 - Choose N as a function of T : $O(T^{2/3})$.
 - Regret: $O(T^{2/3})$.

Algorithm - Main Idea #2

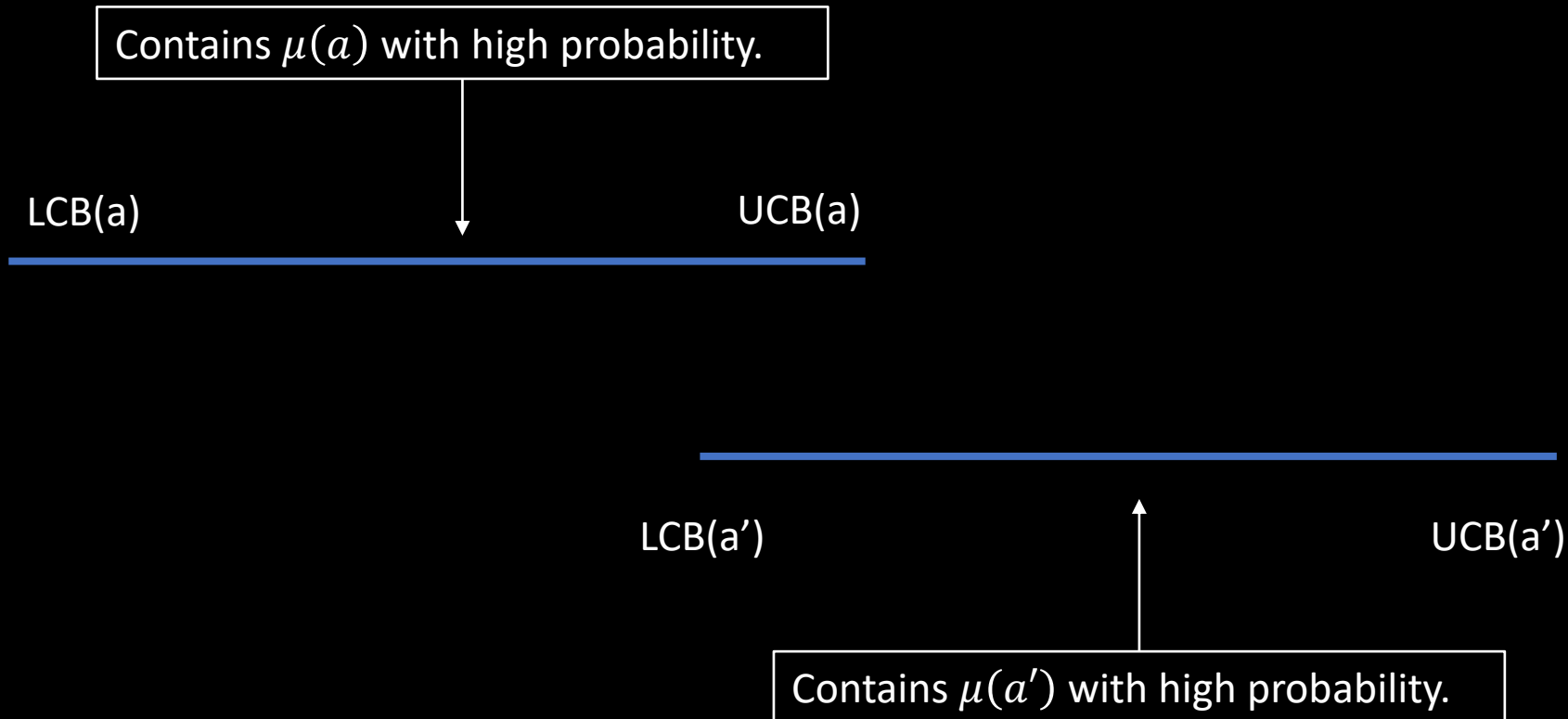
- Adaptive exploration.
- Use UCB/LCB estimates (upper/lower confidence bounds)

$$\bar{\mu}_t(a) \pm \sqrt{\frac{2 \log T}{n_t(a)}}$$



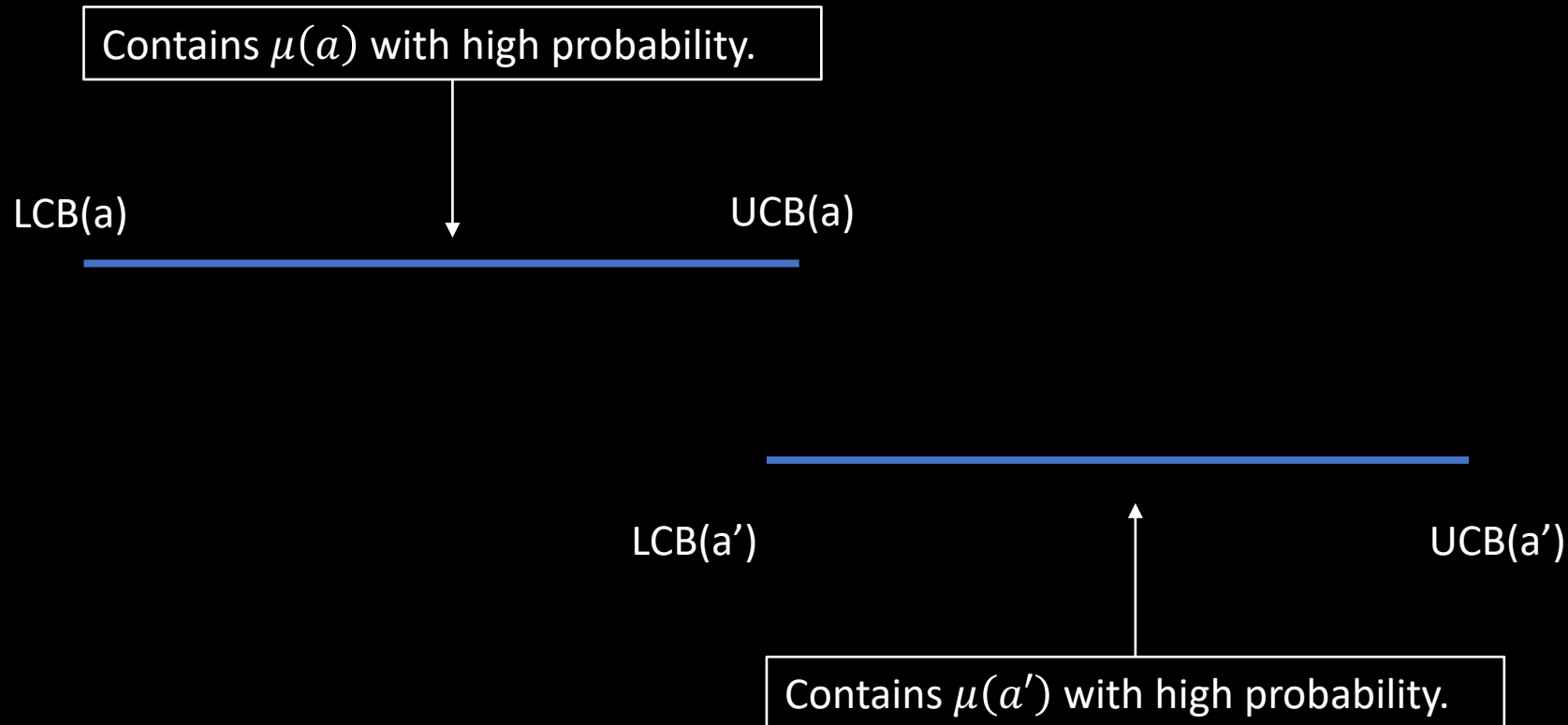
Algorithm - Successive Elimination

- Successive elimination: eliminate obviously suboptimal arms.



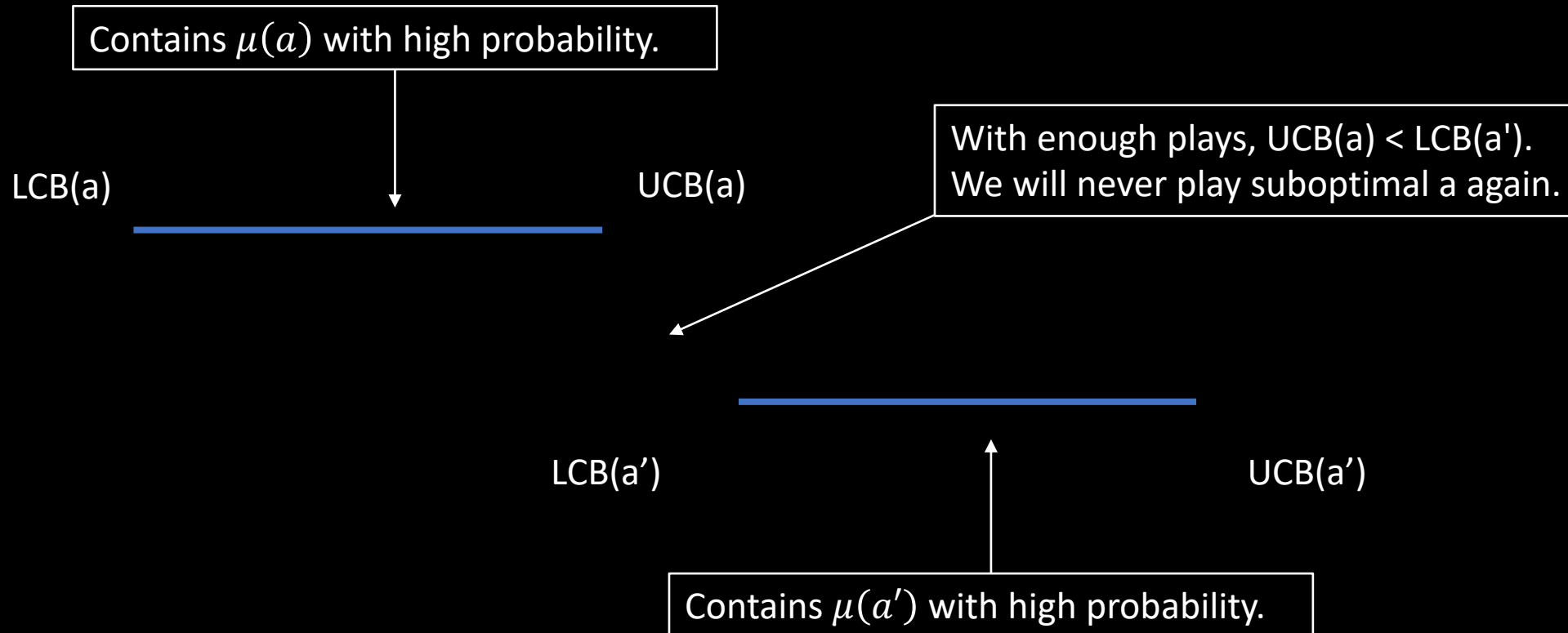
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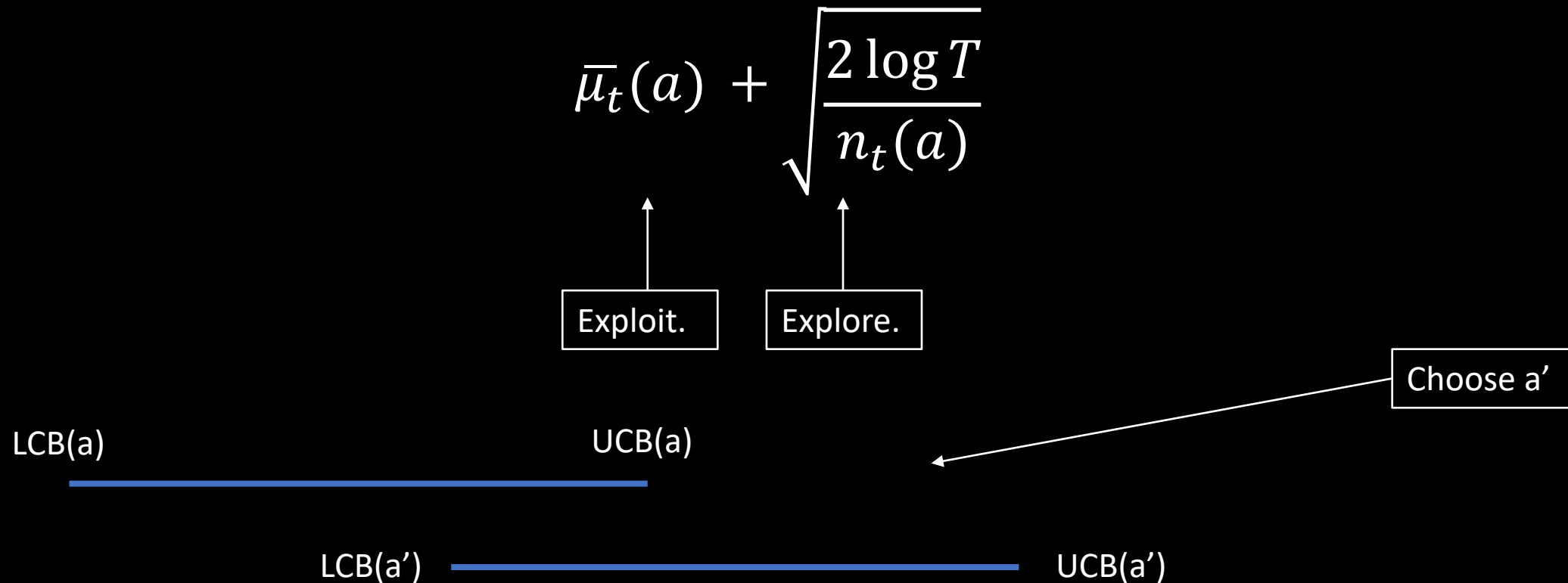
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Algorithm - UCB1

- UCB1: choose arm with highest current UCB.



Extensions

- The basic stochastic bandits model doesn't fully capture reality.
- There are numerous extensions.
 - Algorithms build upon basic ideas introduced today.

Extensions

- Continuous actions.
 - Two arms similar implies their rewards are similar.
 - “Zooming” algorithm – adaptive discretization.

Extensions

- Adversarial feedback.
 - Rewards can be chosen by an adversary.
 - No distributional assumptions.
 - Requires a different algorithm (EXP3).

Extensions

- Bandits with Knapsacks.
 - Finite number of resources with some initial budget.
 - Each pull earns reward and consumes resources.
 - Balance resource consumption while earning reward.

Extensions

- Linear Contextual Bandits.
 - Reward is a linear function of feature vector and unknown weight vector.
 - Find the optimal policy mapping feature vectors to arms.

Extensions

- Delayed feedback.
- Dueling bandits.
- Sleeping bandits.
- Multi-player bandits.
- Bandit convex optimization.
- More...
- Combinations of the previous extensions!
 - Bandits with Knapsacks with continuous actions.
 - Linear Contextual Bandits with Knapsacks.
 - ...

Conclusion

- Multi-armed bandits model online decision-making problems.
 - Decisions must be made one by one.
 - Observe feedback only for the chosen decision.
 - How to choose a good sequence of decisions?
- Very well-studied with numerous extensions.

Thank you!